Graphical Abstract

Laying the Foundation for the Clinical Translation of Deep Learning-based Spectral Fitting Models in Magnetic Resonance Spectroscopy

John T. LaMaster, Kay C. Igwe, Jamie Near, Mauricio Reyes, Roland Kreis, Rudy Rizzo

Highlights

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- Inconsistent and heterogeneous validation protocols used in deep learning-based spectral fitting studies for MR spectroscopy are reviewed and sources of inconsistency are identified.
- Reporting gaps that limit the interpretation of model behavior across SNR, out-of-distribution data, uncertainty, and multivariate metabolite-prediction settings are defined.
- Detailed statistical, agreement, calibration, error, regression, and uncertainty analyses are recommended to define validated model operating ranges for clinical translation.

Laying the Foundation for the Clinical Translation of Deep Learning-based Spectral Fitting Models in Magnetic Resonance Spectroscopy

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ABSTRACT

Magnetic resonance spectroscopy provides a unique window into *in vivo* neurochemistry, enabling non-invasive quantification of metabolite concentrations that carry direct clinical relevance in neurological and psychiatric diseases, aging, oncology, and many other pathological conditions. However, extracting this biochemical information depends on accurate spectral fitting, which remains one of the major bottlenecks in the MRS analysis pipeline. Deep learning (DL) has emerged as a compelling alternative to traditional spectral fitting algorithms, offering the potential for substantial gains in speed, robustness, and scalability. Despite a growing body of published work, however, the field lacks a principled, unified framework for evaluating these models in a manner commensurate with the rigorous standards required for clinical translation. Inconsistent evaluation practices, inappropriate statistical assumptions, poorly characterized data pipelines, and the absence of standardized benchmarking and comparison methodologies have collectively impeded progress toward deployment in clinical settings.

This work addresses that gap directly by presenting a comprehensive methodological framework, grounded in statistical theory, for the rigorous characterization, validation, and benchmarking of DL-based spectral fitting models in MRS. The framework spans the full evaluation pipeline: from data curation and synthetic data considerations for training and evaluation, through sample size analysis and out-of-distribution error characterization, to a curated suite of characterization metrics, uncertainty quantification strategies, and method comparison study designs. Where conventional parametric assumptions are unsuitable for adequately characterizing the output distributions typical of DL models, alternative methods are detailed. All proposed methodologies are motivated through statistical theory.

By establishing a rigorous and accessible foundation, this work aims to accelerate the responsible clinical translation of DL-based spectral fitting, bridging the divide between methodological innovation and real-world deployment in MRS.

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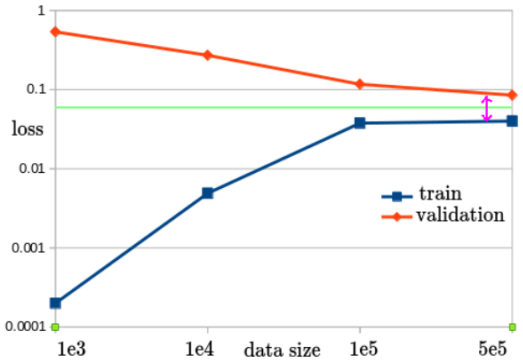


Figure 1: Example of a dataset size analysis from [22]. Beyond a variable threshold, the gains from additional data plateau.

Files: "Hatami_01.png",
Label: fig:dataset size analysis

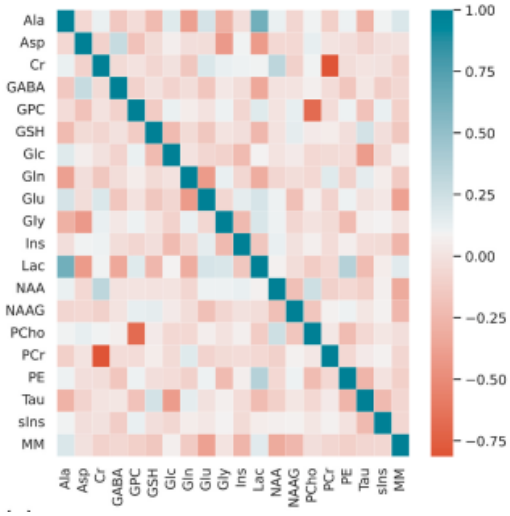


Figure 2: Heatmap used to visualize the correlation analysis of the predicted variables. Unbiased estimators should show minimal correlation between predicted variables. Adapted from Shamaei et al. [50]

File: "Shamaei2023.png",
Label: fig:correlation matrix heatmap

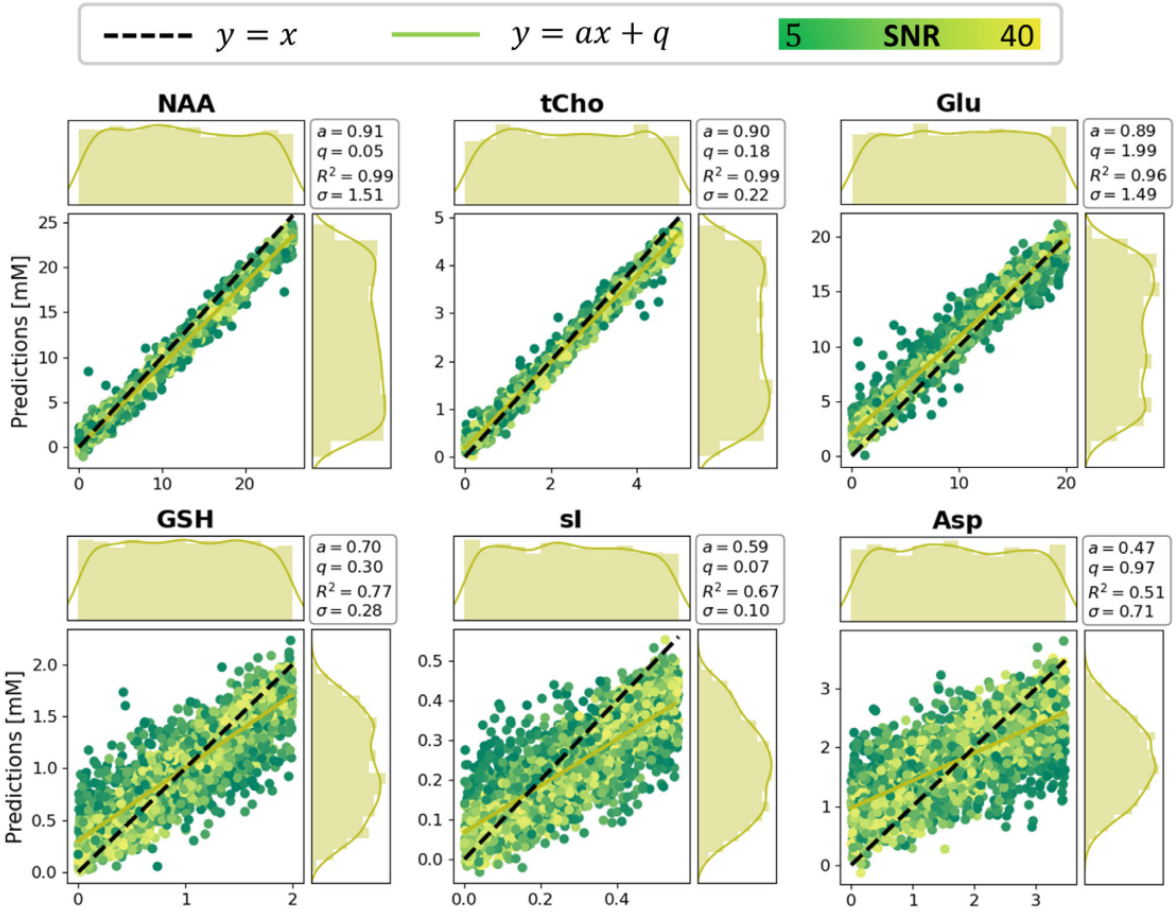


Figure 3: Maps and marginal distributions of selected metabolites modeled by a neural network. The ground truth values, plotted along the x-axis, were sampled from uniform distributions, but the networks predict marginal distributions that range from unimodal Gaussian to approximately uniform distributions. Adapted from Rizzo et al. [46].

File: "Rizzo_01.png",
Label: fig:maps and distributions

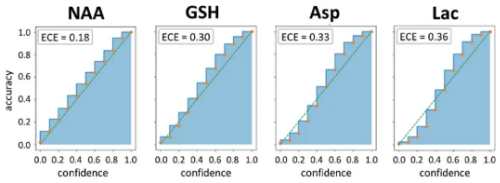


Figure 4: Reliability diagrams with the expected calibration errors for selected metabolites predicted by a neural network. Adapted from Rizzo et al. [47].

File: "Rizzo_02.png",
Label: fig:calibration plots

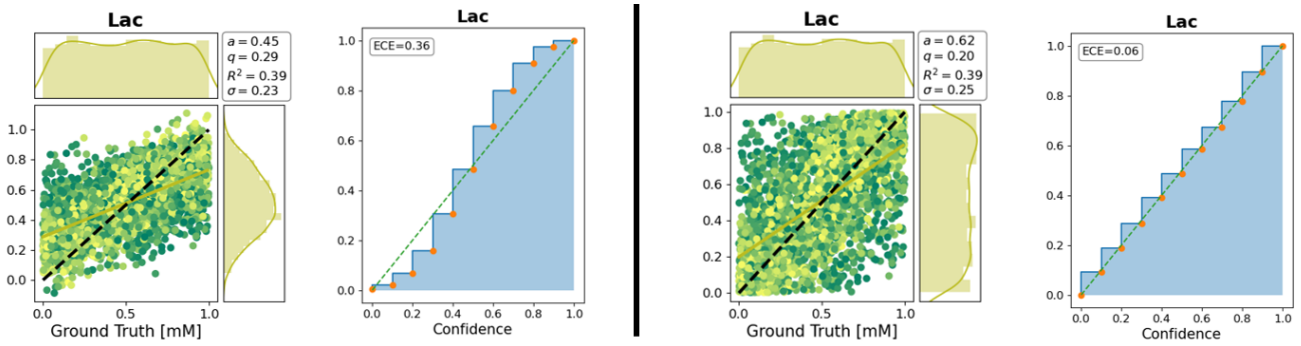


Figure 5: shows an example of pitfalls for naive recalibration for lactate. On the left, the original model results are reported where lactate's predictions are gaussianly skewed to the mean value (y-axis) in correspondence to a uniformly testing set (x-axis). The analysis of its calibration confirms the findings, reporting low concentrations that tend to be overestimated (system is overconfident for confidence < 0.4) and high concentrations that tend to be underestimated (system underconfident for confidence > 0.6). On the right, naively recalibrating leads to a perfect reliability diagram with the predictions that statistically display as uniformly distributed, in agreement with the statics of the ground truth. Unfortunately though, estimates of individual values are far off from the 1:1 relationship between estimate vs. ground truth and are instead distributed across the labeled concentration range to yield a correct uniform distribution of predicted values. Figure adapted from citation.

File: "Rizzo_03.png",
Label: fig:recalibration

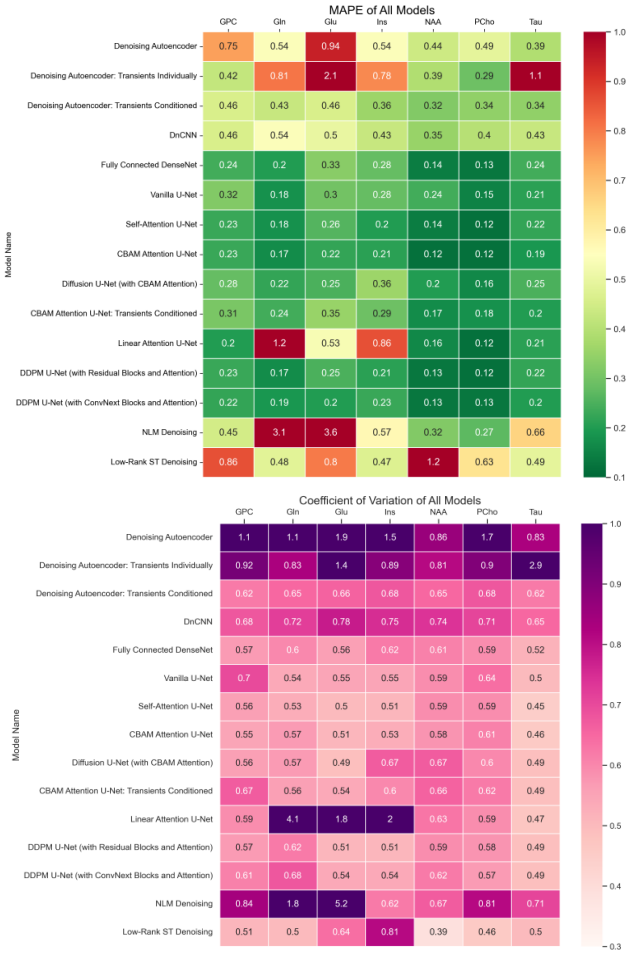
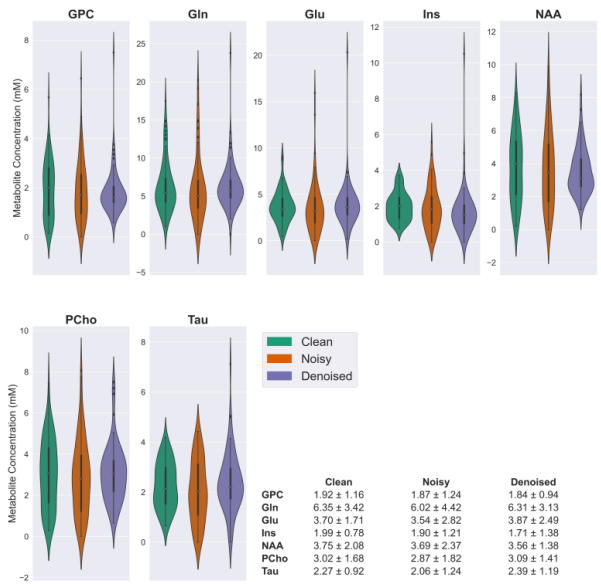


Figure 4.99: Heatmap showing the coefficients of variation for denoised data from the coil-combined dataset for all the models. The coefficients of variation were calculated as the standard deviation divided by the mean values of quantification for each metabolite.

Figure 6: Heatmaps showing visualizations of comparison metrics for multiple variables across a large number of models. Adapted from Gupta et al. [20].

Files: "Gupta_01.png", "Gupta_02.png"
Label: fig:comparing heatmaps



	Clean	Noisy	Denoised
GPC	1.92 ± 1.16	1.87 ± 1.24	1.84 ± 0.94
Gln	6.35 ± 3.42	6.02 ± 4.42	6.31 ± 3.13
Glu	3.70 ± 1.71	3.54 ± 2.82	3.87 ± 2.49
Ins	1.99 ± 0.78	1.90 ± 1.21	1.71 ± 1.38
NAA	3.75 ± 2.08	3.69 ± 2.37	3.56 ± 1.38
PCho	3.02 ± 1.68	2.87 ± 1.82	3.09 ± 1.41
Tau	2.27 ± 0.92	2.06 ± 1.24	2.39 ± 1.19

Figure 7: Violin plots depicting box-and-whisker plots and the shape of the distributions to visualize how a model (purple) captures the underlying target distribution (green) compared to the reference method (orange). Adapted from Gupta et al. [20]. Will replace with updated version from Jin X.'s thesis (2024)

File: "Gupta_03.png"
Label: fig:violin plots

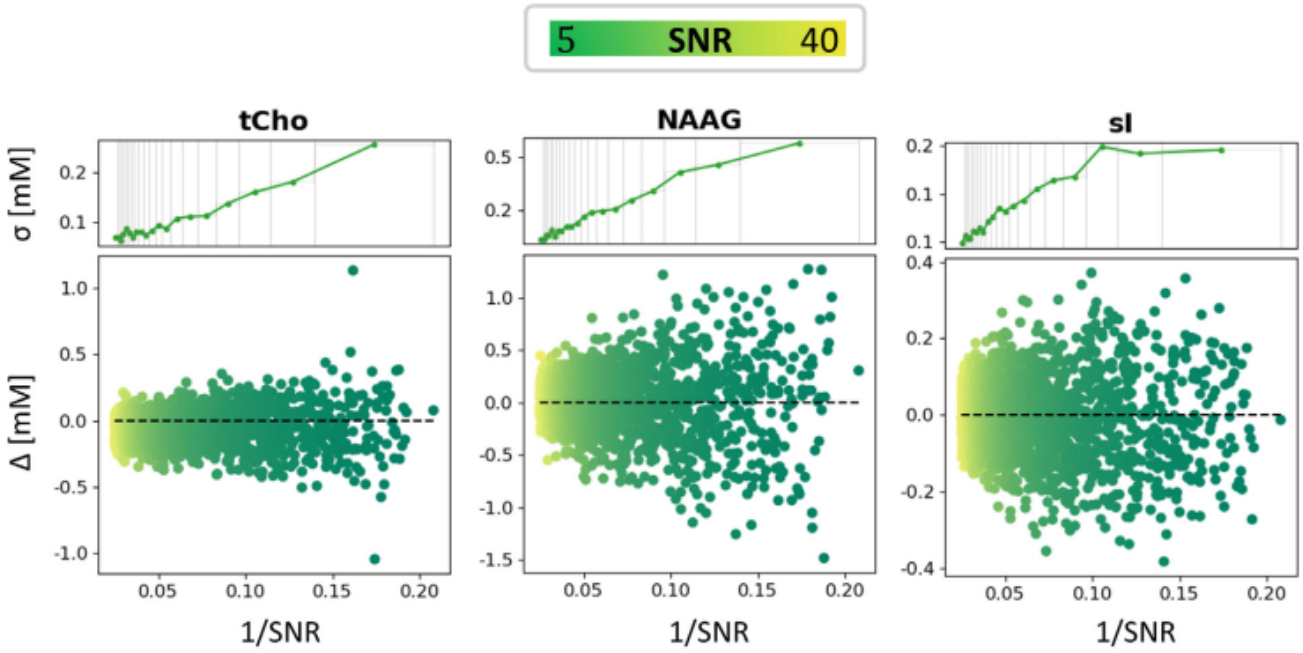


Figure 8: Metabolite quantification errors Δ and the RMSE σ as a function of the inverse spectral SNR. Lower SNR values yield wider error dispersions. Adapted from Rizzo et al. [46]

File: "Rizzo_04.png"
Label: fig:SNR dependence

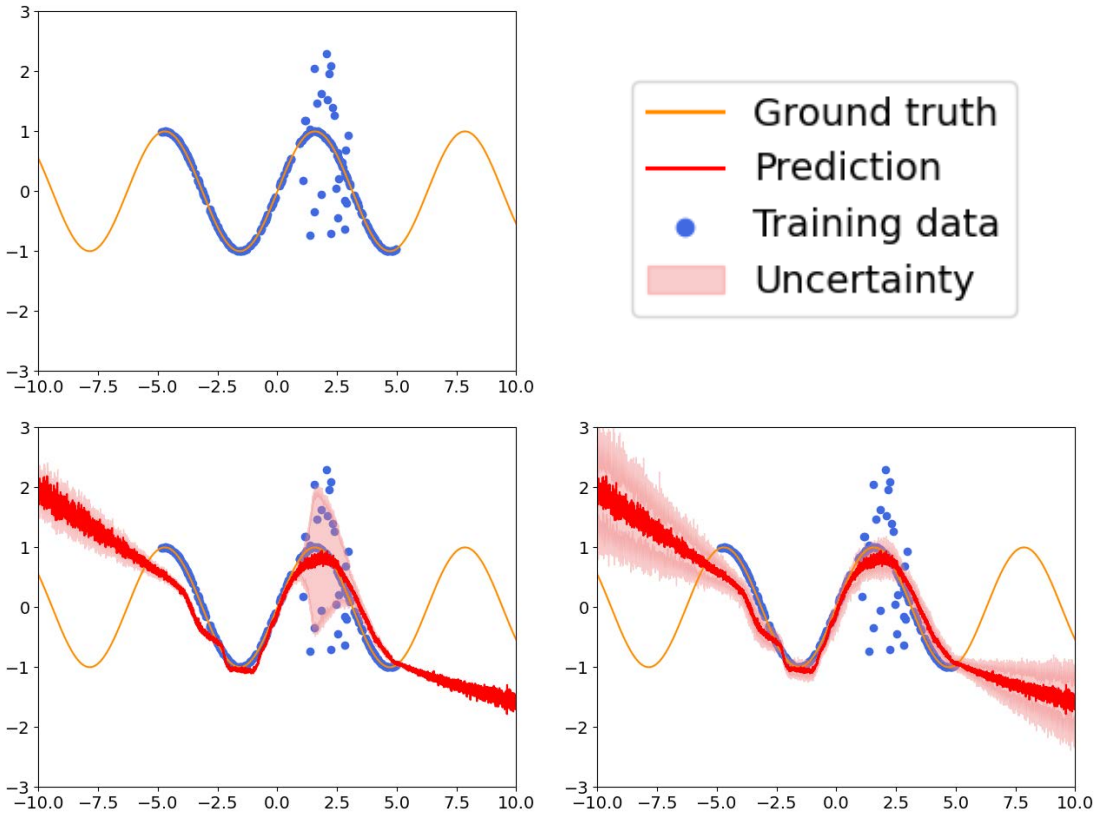


Figure 9: (Top) A toy example of a data set (training set, blue) is distributed in a 1D domain D = [−4, +4] following a mostly sinusoidal pattern that matches ground truth (yellow, extended to a bigger domain D-GT- defined in [-8,8]). A DL model (not displayed) predicts the sinusoidal wave (red). Good performances are reported within the training data domain D and noise divergent from the sinusoidal way characteristic is seen out of the training data domain. Visualization of aleatoric (bottom-left) and epistemic (bottom-right) uncertainties is reported in shade of red. Despite perfect predictions, a measure of uncertainty relates to the interval [1, 3] where training data suffered a bigger variance (aleatoric). A second measure of uncertainty relates to training distribution, displaying higher uncertainty when predictions are asked for data outside the training domain D (epistemic). Modified from Xu et al. [62]

File: "toy_example2.pdf"
Label: fig:uncertainty toy example

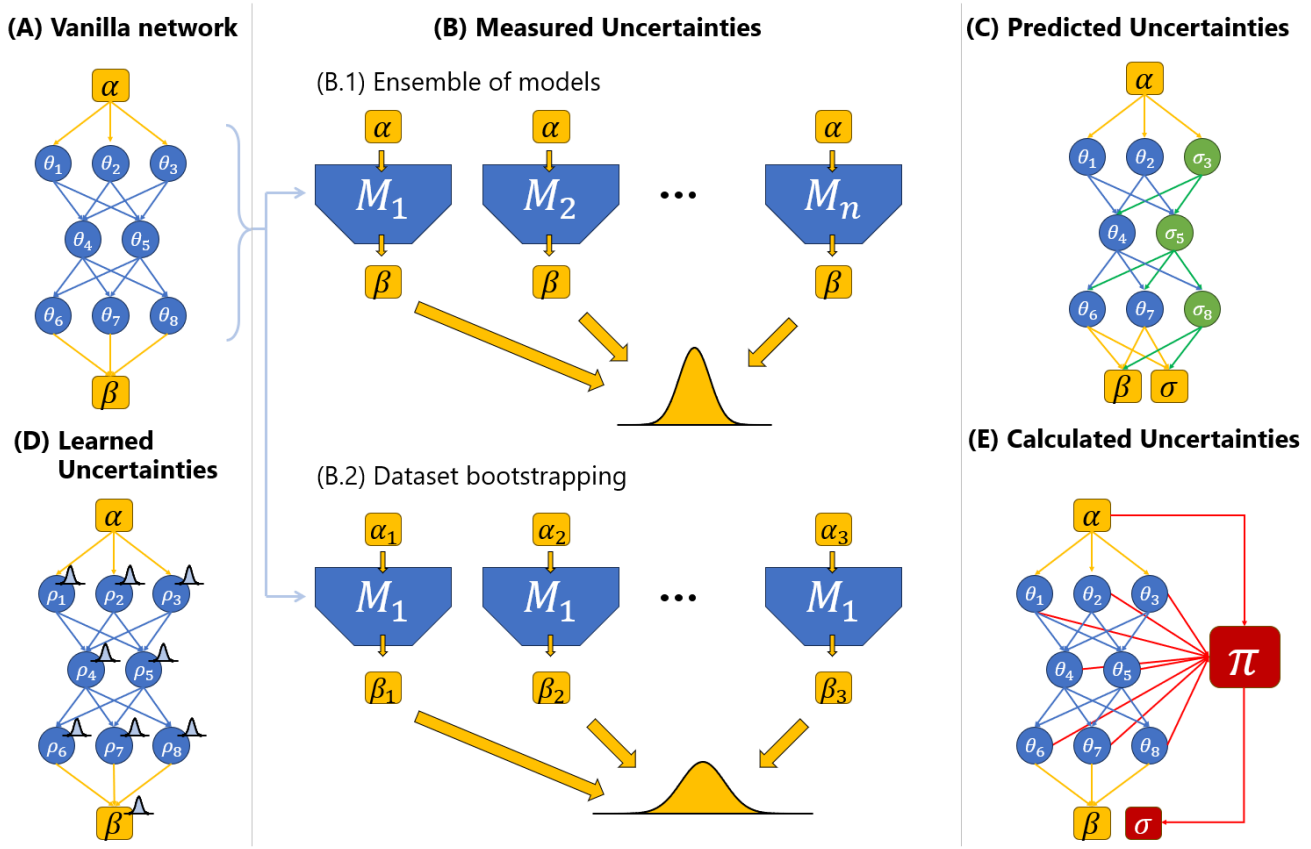


Figure 10: Diagram representing various estimators for uncertainties in DL. (A) Vanilla DL network, alpha input, Beta output and θ_i are different deterministic network nodes. (B) Measured Uncertainties. B.1 Multiple pre-trained models either on similar or different dataset are tested for the same testset alpha and produce different predictions Beta. B.2 The same pretrained network M1 is tested on different test sets producing different output Beta. For both cases, the evaluation of the single model predicted output as an aggregate gives a measure of uncertainty reflected in the estimate's distribution. (C) Predicted uncertainties include specific nodes in the network that are dedicated to embed features reflected as output into a measure of the output variability (sigma). (D) Learned uncertainties shift the network paradox introducing stochastic nodes (rho) instead of deterministic ones (Theta). A stochastic node activation varies according to a distribution. The output will inherit stochastic properties from the network. (E) Calculated uncertainties relies on the re-definition of part of the network by encoding a specific physic-driven set of nodes (pi). The mathematical model definition defines a proxy to Fisher information calculation and generates inference to CRLBs.

File: "ours_02.png"
Label: fig:uncertainty estimation types

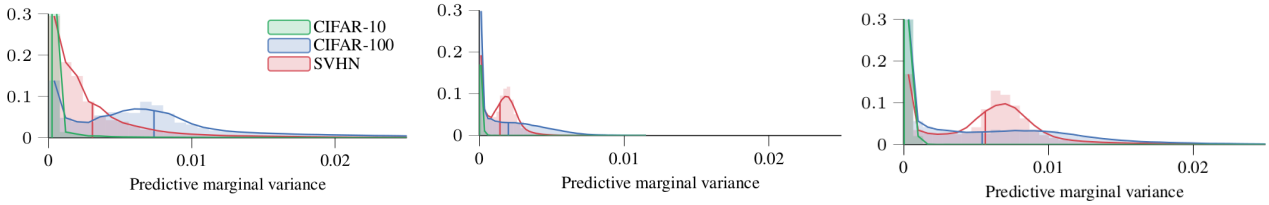


Figure 11: Histograms of sample-level variances of different datasets. From left to right, these plots show the OOD detection capability of models with increasingly well-calibrated uncertainty predictions. Here, green is the in-distribution dataset, blue is in the middle of ID and OOD, and red is strongly OOD. Adapted from Meronen et al. [39].

Files: "Meronen_01.png", "Meronen_02.png", "Meronen_03.png"

Label: fig:predictive marginal variance

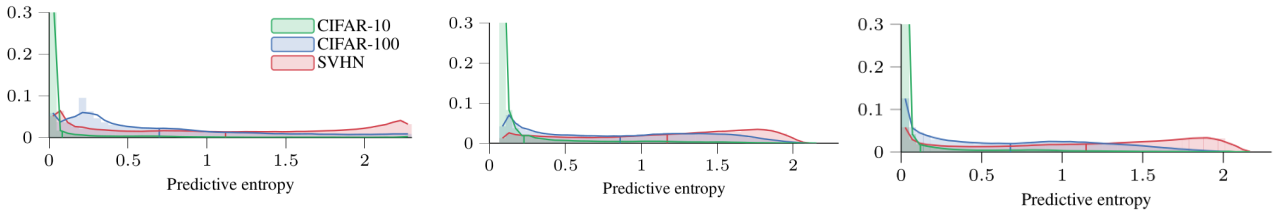


Figure 12: Histograms of the entropies of the mean predicted values. The models correspond to those in Fig. 11 above. Here, green is the in-distribution dataset, blue is in the middle of ID and OOD, and red is strongly OOD. Adapted from Meronen et al. [39].

Files: ”Meronen_04.png”, ”Meronen_05.png”, ”Meronen_06.png”

Label: fig:predictive entropy

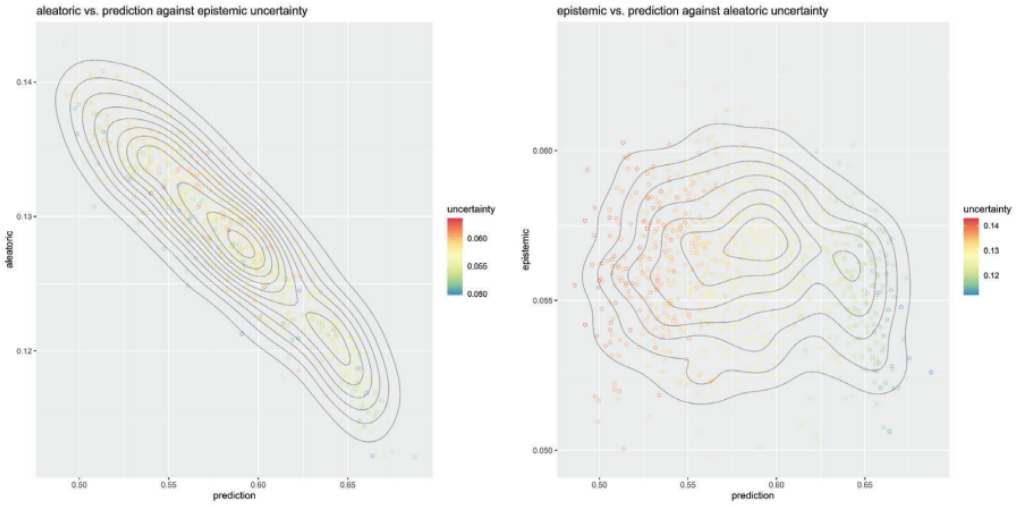


Figure 13: Expected patterns of aleatoric and epistemic uncertainty. Accuracy increases as aleatoric uncertainty decreases (shown on the left) while the epistemic uncertainty has no discernible pattern. On the right, the epistemic uncertainty still has no pattern, but the aleatoric uncertainty decreases from left to right as the prediction accuracy increases. Adapted from Ghoshal et al. [13].

File: "Goshal_01.png"
Label: fig:decomposed uncertainties

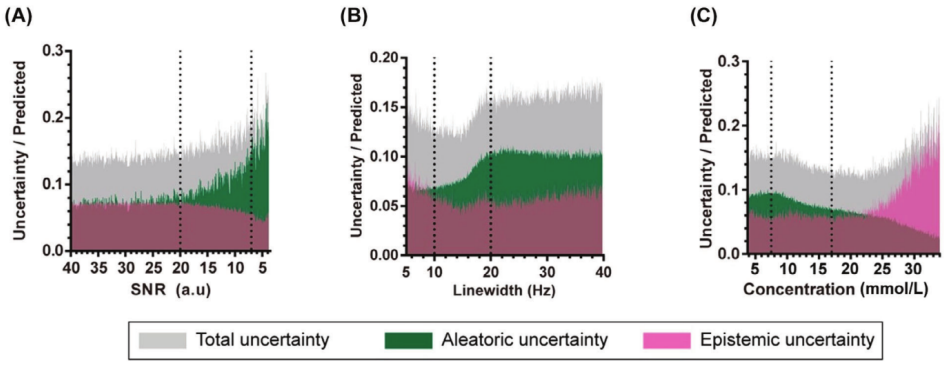


FIGURE 6 Aleatoric (green), epistemic (pink), and total (gray; aleatoric + epistemic) uncertainty of NAA as a function of (A) SNR, (B) linewidth, or (C) concentration using the BCNN and the simulated data set II. The BCNN was trained on the simulated data set I, of which the ranges of SNR, linewidth, and concentrations of NAA are indicated with the dotted vertical lines. The amount of uncertainty was obtained by calculating the area of the NAA signal at ~2.0 ppm from each of the 3 types of the uncertainty spectra and normalizing it to that from the respective BCNN-predicted (metabolite-only) spectrum. Note that in (A) the SNR is decreasing (from left to right) along the *x*-axis

Figure 14: Breakdown of the aleatoric, epistemic, and total uncertainy for individual predicted variables across their parameter spaces highlighting the heteroscedasticity. Adapted from Lee and Kim [31]

File: "Lee_01.png"
Label: fig:component-level uncertainty

Declaration of competing interest

The authors have no competing interests to declare.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process.

During the preparation of this work, the authors used GPT-5 models to support the exploration and understanding of methodological approaches drawn from interdisciplinary fields and in considering their applicability to the context of this work. The models were also used for language refinement, editorial revision, and improving the clarity and presentation of the manuscript. During and after using this tool/service, the authors reviewed, edited, and verified the content as needed and take full responsibility for the content of the published article.

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CRediT authorship contribution statement

John T. LaMaster: Conceptualization, Investigation, Methodology, Visualization, Writing - Original Draft, Writing - Review & Editing. **Kay C. Igwe:** Conceptualization, Investigation, Methodology, Visualization, Writing - Original Draft, Writing - Review & Editing. **Jamie Near:** Conceptualization, Investigation, Methodology, Writing - Review & Editing. **Mauricio Reyes:** Investigation, Writing - Review & Editing. **Roland Kreis:** Conceptualization, Investigation, Methodology, Writing - Review & Editing. **Rudy Rizzo:** Conceptualization, Investigation, Methodology, Visualization, Writing - Original Draft, Writing - Review & Editing.

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A. Deep Learning Definitions

Interdisciplinary fields must contend with terminology barriers between their constituent disciplines. Deep learning researchers working in spectroscopy and spectroscopists working with deep learning are well aware of these differences: some terms carry different meanings across fields while others simply do not exist in one of them. This work addresses researchers in both fields who have varying levels of familiarity with the other; the following definitions are intended to minimize terminological confusion.

A priori Refers to knowledge or constraints that are known or assumed before training begins and that are not updated by the data. In the context of deep learning for MRS, *a priori* information includes physical relationships (e.g., the known lineshape model or the equation for zero-order phase correction) that are embedded directly in the model architecture or loss function rather than learned from data. This is in contrast with *prior* (see below), which specifically denotes a probability distribution over model parameters in a Bayesian formulation.

Ablation, data Data ablation refers to the deliberate, randomized omission of features when constructing a synthetic training dataset. The motivation is to expose the network to examples in which a given feature is absent, supplementing its learned representation and preventing it from learning undesirable correlations or becoming dependent on any feature always being present.

Ablation, study A systematic experimental procedure common in deep learning in which a baseline architecture is progressively modified to isolate the contribution of each proposed component to observed performance improvements. Components are removed one at a time (or in combination) to determine which are necessary and which are redundant.

Activation function A nonlinear scalar function applied element-wise to the output of a linear operation (weighted sum or convolution) within a neural network layer. Without activation functions, stacked linear operations would collapse to a single linear transformation regardless of depth. The most widely used activation function is the rectified linear unit (ReLU) [42]:

$$\sigma(x) = \max(0,\, x), \tag{1}$$

which zeros out negative values. Other common choices include Leaky ReLU, ELU, GELU, Sigmoid, and Softmax, each offering different gradient and saturation properties. The output of the activation function is passed to the subsequent layer or as the final output.

Augmentation, data The process of applying label-preserving transformations to training samples to artificially increase the effective size and diversity of the training set, thereby improving generalization [52]. For images, common augmentations include affine transformations, color and contrast manipulations, and histogram normalization. For MRS, relevant augmentations include zero- and first-order phase offsets, frequency shifts, eddy current distortions, baseline variations, residual water signals, macromolecular contributions, and noise scaling.

Baseline (model) A model whose performance serves as a reference point for evaluating the proposed work. Baselines may be simple heuristics, conventional non-DL methods (e.g., LCModel, AMARES), or prior DL architectures. Meaningful performance comparisons require that baselines be evaluated under identical conditions.

Batch size The number of training samples processed in a single forward-backward pass before the model weights are updated. Larger batch sizes produce more stable gradient estimates but require more memory and may converge to sharper, less generalizable minima [27]. Smaller batches introduce noise that can act as implicit regularization.

Bias The term carries two related but distinct meanings. In a neural network layer, bias b is a learnable scalar parameter added to the weighted sum of inputs before the activation function:

$$f(\mathbf{x}) = \sigma\left(\sum_{i=1}^{n} x_i W_i + b\right), \tag{2}$$

allowing the activation to shift independently of the input magnitude. Refer to Appendix B for the statistical definition.

Class balance The condition in which all categories or subgroups of interest are equally represented in a dataset. Curating a class-balanced dataset prevents the model from developing a prior bias toward more frequently occurring classes during training.

Class imbalance The condition in which categories or subgroups are unequally represented in a dataset, causing the model to be implicitly biased toward the majority class. Class imbalance can be addressed through oversampling of minority classes, undersampling of majority classes, or loss function weighting.

Data, simulated Data generated by a computational forward model based on known physical principles, with parameters drawn from a defined distribution. In MRS, simulated data are produced using quantum mechanical density matrix simulations that propagate the spin system through the pulse sequence, yielding theoretically exact metabolite basis functions to which noise, lineshape distortions, and other spectral components and artifacts are then applied. Because the generative process is fully specified, ground truth labels are exact and available in unlimited quantities.

Data, synthetic Data generated by any computational means to supplement or replace experimentally acquired data, encompassing both physics-based simulation and purely statistical generative models such as GANs, VAEs, and diffusion models. Simulated data (see above) is a strict subset of synthetic data. The broader term also includes data generated by learned or empirical models. The two terms are sometimes used interchangeably in the MRS literature, though the distinction is meaningful when the fidelity of the generative process is relevant.

Distribution In the context of training and evaluation, a distribution refers to the full set of statistical characteristics of a dataset, including feature ranges, noise levels, class frequencies, and acquisition conditions, rather than the data samples themselves. Two datasets drawn from the same distribution are expected to be statistically interchangeable for the purposes of model training and evaluation.

DL quantification A class of DL models trained exclusively to predict metabolite concentrations, bypassing the estimation of intermediate spectral modeling parameters such as linewidth, phase, and spectral baselines. This approach is unique to deep learning and has no direct analog in conventional spectral fitting.

DL spectral modeling A class of DL models that predict the full set of spectral modeling parameters analogous to those estimated by conventional spectral fitting algorithms, including metabolite concentrations, lineshape parameters, phase offsets, baseline components, and other nuisance terms. These models most closely mirror traditional approaches such as LCModel or Osprey and enable direct parameter-level comparisons.

Domain adaptation A transfer learning strategy that adapts a model trained on a source domain to perform well on a target domain with a different but related distribution, typically with limited or no labeled target data [60]. In MRS, relevant domain shifts include changes in scanner vendor, field strength, pulse sequence, echo time, and tissue type.

Domain shift The statistical discrepancy between the distribution of training data (source domain) and the distribution of data encountered at inference time (target domain) [60]. Sometimes these are referred to as training and deployment distributions, respectively. Domain shift is the primary cause of performance degradation when models trained on synthetic or scanner-specific data are applied in new clinical environments. Quantifying and mitigating domain shift is a central challenge in the clinical translation of DL-based MRS methods.

Dropout A regularization technique in which, during each training step, individual neurons or structured groups of neurons are randomly set to a constant, generally zero but sometimes α, with the dropout rate of probability p [56]. Naïve (unstructured) dropout zeroes individual elements of the activation tensor. Structured dropout zeroes out correlated units such as entire channels or spatial locations. At inference, dropout is typically disabled and weights are scaled by $(1 - p)$. When dropout is retained at inference time and applied across multiple forward passes, the resulting ensemble of stochastic predictions approximates Bayesian posterior sampling (Monte Carlo dropout) [12].

Embedding / feature representation A dense, continuous vector representation of an input computed by the intermediate layers of a neural network. The embedding encodes the learned abstract structure of the input in a lower-dimensional space that is more useful for the target task than the raw input. In spectral fitting architectures, the embedding produced by the encoder captures spectral features relevant to the estimation of fitting parameters, or other parameters of interest.

Error The scalar or vector deviation of a predicted value $\hat{y}$ from the corresponding ground truth y: $e = \hat{y} - y$. Signed error distinguishes over-estimation from under-estimation. Various aggregate scalar metrics (RMSE, MAPE, etc.) summarize the error distribution across a mini-batch or dataset.

Forward model A mathematical description of the physical process that maps model parameters to the observable data. In MRS, the forward model maps metabolite concentrations, lineshape parameters, phase, and baseline coefficients to a reconstructed spectrum. Forward models are used to generate simulated training data and, in physics-constrained architectures, to enforce consistency between network predictions and the measured signal during training or inference.

Generalizability The capacity of a trained model to produce accurate predictions on data that was not seen during training. A model is said to generalize well if its performance on held-out data approaches its performance on its training data. Generalizability exists on a spectrum relative to the degree of distributional shift: a model is generally expected to perform best on in-distribution data, somewhat worse on mildly out-of-distribution data, and worst on strongly out-of-distribution data. The more a model generalizes, the more out-of-distribution data it can accurately predict, which creates a design tension: models should generalize to data that is slightly to moderately out-of-distribution but are not expected to extrapolate reliably to strongly out-of-distribution inputs. The boundaries between mild, moderate, and strong distribution shift cannot be universally defined and must be characterized empirically [34].

Ground truth The target output against which network predictions are compared during loss computations. In supervised learning for MRS spectral fitting, the ground truth consists of the known simulation parameters (concentrations, lineshape terms, phase offsets, etc.) used to simulate the synthetic training spectra. The quality and accuracy of the ground truth directly bounds the performance of any supervised model trained on it.

Hold-out set A subset of the dataset that is partitioned before any model training or hyperparameter optimization and reserved exclusively for final performance evaluation. Unlike the validation set, which may influence model selection or early stopping, the hold-out set must not influence any training decision. Exposure of the hold-out set during development constitutes data leakage and invalidates the performance estimates. This is commonly referred to as the *test set*.

In-distribution (ID) data Data that is statistically consistent with the training distribution, having been generated by the same acquisition protocol, simulation pipeline, or experimental conditions. Models are expected to perform best on in-distribution data because the feature space should be well covered by training examples [23].

Inverse problem The problem of inferring the parameters of a generative model from observed data, i.e., recovering the cause from the effect. In MRS, the forward problem maps metabolite parameters to a spectrum. The inverse problem recovers the parameters from the measured spectrum. Inverse problems are often ill-posed in the Hadamard sense: a solution may not exist, may not be unique, or may not depend continuously on the data [4]. Deep learning methods for spectral fitting are fundamentally inverse problem solvers and inherit this ill-posedness, which motivates the use of regularization and uncertainty quantification.

Labels / annotations Synonymous terms for the target values associated with each input sample in a supervised learning dataset. For image classification, labels denote the object class. For regression tasks such as spectral fitting, labels are the ground truth parameter values the network is trained to predict. Labels are distinct from the input data itself: the input spectra are not labels, but the simulation parameters used to simulate them are.

Last-layer The final layer of a deep learning architecture, typically a linear (fully connected) layer that maps the learned feature representation to the output parameter space. The term became prominent with Bayesian last-layer (BLL) networks [30, 55], in which the backbone is trained as a standard point-estimate model while only

the last layer is treated as Bayesian, i.e., its weights are represented by learned mean and variance parameters and sampled at inference to produce a predictive distribution. This formulation substantially reduces the computational cost of full Bayesian inference.

Loss A scalar value computed from the network predictions and the ground truth labels that quantifies the model's error for a given batch of data. During training, the gradient of the loss with respect to all learnable parameters is computed via backpropagation and used to update those parameters by gradient descent. A lower loss indicates better agreement between predictions and ground truth for the chosen metric.

Loss function The function used to compute the loss value. The two most widely used loss functions are the mean absolute error (MAE, $L1$ loss) and the mean squared error (MSE, $L2$ loss). The latter penalizes large deviations more severely than the former due to the quadratic term. The only strict mathematical requirement of a loss function is differentiability with respect to the network outputs, which is necessary for gradient-based optimization. Task-specific or physics-informed loss functions are an active area of development in applied deep learning [16]. In practice, loss functions are always minimized. Functions that need to be maximized (like the log-likelihood) are made negative (negative log-likelihood (NLL)) and then minimized.

Model vs. network Although often used interchangeably, these terms carry a useful distinction. A *network* refers specifically to the parameterized estimator defined by the architecture, i.e., the computational graph whose weights are optimized during training. The *model* is the broader system that encompasses the network alongside any auxiliary components such as physics-based forward models, pre- and post-processing transformations, or uncertainty estimation modules. For example, in a deep spectral fitting system that predicts parameters and reconstructs the fit, the network predicts the parameters while the model additionally includes the spectral reconstruction step.

Out-of-distribution (OOD) data Data that differs from the training distribution in one or more statistical or physical properties [23]. In MRS, OOD scenarios include different pulse sequences (e.g., STEAM vs. PRESS), echo times, field strengths, vendor-specific implementations, SNR ranges, pathological tissue types, and cohort-specific distributions not represented during training. OOD data is expected to degrade model performance, but the degree of degradation depends on the magnitude of the distributional shift.

Parameter estimation step The final computational stage in which the encoded feature representation is mapped to the output parameter values. In most DL architectures this consists of one or more linear (fully connected) layers. A multi-layer stack is referred to as a multi-layer perceptron (MLP) head. Depending on the framing, this term may refer narrowly to the last linear layer alone or more broadly to the entire MLP head.

Posterior The posterior probability distribution is the probability distribution over the model parameters θ, given the observed data $\mathcal{D}$, obtained by applying Bayes' theorem [35]:

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta)\, p(\theta)}{p(\mathcal{D})}, \tag{3}$$

where $p(\mathcal{D} \mid \theta)$ is the likelihood, $p(\theta)$ is the prior, and $p(\mathcal{D})$ is the marginal likelihood (model evidence). The posterior summarizes all information about the parameters after observing the data. Predictions can then be obtained by marginalizing over $\boldsymbol{\theta}$ with respect to the posterior, yielding the posterior predictive distribution.

Prior The prior probability distribution $P(\theta)$ encodes beliefs about the model parameters before any data are observed. During training, the prior is updated by the likelihood of the observed data to yield the posterior (see above). The prior can encode domain knowledge, for example by penalizing implausible parameter values, and act as a regularizer in the maximum a posteriori (MAP) estimate [35]. In contrast with the term *a priori* (see above), which refers more broadly to any knowledge or constraint incorporated into the model architecture before training, not necessarily in the form of a probability distribution.

Quantification model An umbrella category for deep learning models that estimate metabolite concentrations, comprising two primary subgroups: (1) metabolite-only models that predict concentrations directly without estimating intermediate spectral fitting parameters, and (2) deep spectral fitting models that estimate the full parameter set analogous to conventional fitting algorithms.

Self-supervised learning A training paradigm in which pseudo-labels are derived from the data itself, without manual annotation [25]. In one formulation, a preliminary module processes the input and generates a label that supervises the remainder of the network. In contrastive learning, two augmented views of the same sample are labeled "similar" while views from different samples are labeled "dissimilar". The network learns representations that reflect this structure without any externally provided labels.

Stationarity The property of a stochastic process or model whose statistical attributes (mean, variance, autocorrelation) remain constant across its domain. A stationary process is predictable and stable in the sense that its behavior does not depend on absolute position within the domain.

Stationarity, global A process or model whose statistical properties are uniform across the entire domain, meaning that the model has homoscedastic marginal variance (not residuals) and the same statistical description holds across its entire parameter space.

Stationarity, local A process or model whose statistical properties are consistent only within localized regions of the domain, allowing the overall process to be non-stationary while remaining approximately stationary within subregions.

Supervised learning The standard training paradigm in which each input sample is paired with a ground truth label, and the loss is computed as a function of the discrepancy between the network predictions and those labels [16]. The network parameters are updated to minimize this loss over the training set via gradient descent.

Training / validation / test split The partition of a dataset into three non-overlapping subsets, also referred to as *data splits*. The *training set* is used to optimize model parameters by gradient descent. The *validation set* is used to monitor generalization during training, tune hyperparameters, and select among model variants. It cannot be used to compute the final reported performance. The *test*

set (hold-out set) is used exclusively for final performance evaluation and must not influence any training or development decisions.

Uncertainty A quantification of the confidence associated with a model's prediction. Uncertainty is commonly represented as a predictive variance, a credible interval, or an entropy value. The term is sometimes used interchangeably with variance, which is appropriate under Gaussian predictive distributions but may be misleading for multimodal or heavy-tailed distributions.

Uncertainty, aleatoric Uncertainty arising from irreducible stochasticity in the data-generating process, such as measurement noise or intrinsic sample variability [26]. Because it is a property of the data rather than the model, aleatoric uncertainty cannot be reduced by collecting additional training samples. In MRS, noise level and spectral artifacts are primary sources of aleatoric uncertainty.

Uncertainty, epistemic Uncertainty arising from incomplete knowledge of the model parameters, which can in principle be reduced by providing either more or more diverse training data [26]. In practice, epistemic uncertainty is elevated for out-of-distribution inputs where the model lacks relevant training examples. This is in contrast with aleatoric uncertainty, which is irreducible.

Uncertainty calibration A model's uncertainty estimates are said to be calibrated if the predicted confidence levels match the empirical frequencies of correctness, meaning that the predictions' assigned confidence p are correct approximately $p \times 100\%$ of the time [19]. Calibration is typically assessed using reliability diagrams and summarized by the expected calibration error (ECE). A model can be accurate but poorly calibrated (overconfident or underconfident). Calibration post-processing methods, such as temperature scaling, can reduce miscalibration without changing the model's point predictions.

Uncertainty validation The process of empirically verifying that the model's uncertainty estimates are informative and meaningful for the task at hand. Common validation strategies include confirming that high-confidence predictions have lower error (sparsification curves), that uncertainty increases for out-of-distribution inputs, and that predictive intervals achieve their nominal coverage [26].

Unsupervised learning A training paradigm in which no internal or external labels have any meaningful impact on network training. The loss is computed directly between the model outputs and the inputs, or a transformation of them [16]. The model learns to capture intrinsic structure in the data, such as a compressed low-dimensional representations via autoencoders or a generative model of the data distribution.

Vanilla An adjective describing the basic, unmodified form of an architecture or training procedure, used as a reference baseline against which proposed modifications are compared. For example, a vanilla transformer refers to the original encoder-decoder architecture of Vaswani et al. [58] without any task-specific adaptations.

Variance Characterizes the spread of a distribution. In the context of deep learning predictions, variance is a common proxy for uncertainty, though it fully characterizes uncertainty only under Gaussian assumptions. Refer to Appendix B for the statistical definition.

B. Metrics & Statistical Definitions

The definitions provided below are widely available in textbook and research literature. These terms were selected based on their relevance to the topics discussed in this work. Explanations and accompanying equations, where applicable, are presented here to provide a singular resource to help help understand the main text and that can be used as a quick reference.

Agreement Agreement evaluates whether two different methods or instruments yield the same absolute values for the same subject. The two variables must be measured on the same scale and in the same unit. The ideal relationship is a strict $1:1$ identity.

Association Association evaluates whether two variables move together in a predictable pattern, often measured by correlation. The two variables do not need to have the same values, nor do they need to be measured in the same unit.

Bias The term carries two related but distinct meanings. In statistical estimation, bias is the systematic component of error: the expected deviation of an estimator from the true parameter value, $\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$ [59]. An estimator is unbiased if this quantity is zero. In empirical method comparison, systematic bias is estimated from the location of the distribution of pairwise differences, using, for example, the mean or a robust median-based estimator. In regression, a non-zero intercept is a canonical example of additive bias. Refer to Appendix A for the deep learning definition.

Bias, conditional Conditional bias, also known as random or probabilistic bias, is the systematic error of an estimator at a specified value or set of conditions. For a probabilistic predictor, it can be expressed as the difference between the expected prediction and the corresponding target value $\mathbb{E}[\hat{y} \mid x] \neq y$. It remains a form of estimator bias and is distinct from the random variability represented by the predictive distribution.

Bias, constant Constant bias is an additive systematic error that has the same magnitude and direction across the measurement range. It shifts all estimates by approximately the same amount, independent of the magnitude of the reference measurement. In the linear regression model $x = my+b$, constant bias is represented by a non-zero intercept b when the ideal identity relationship is $x = y$. In this work, the term is used primarily in method-comparison analyses to describe a systematic additive disagreement between the candidate and reference methods that is independent of measurement magnitude.

Bias, fixed Fixed bias is a systematic additive error that does not depend on the magnitude of the target value and is represented by a non-zero regression intercept. In this work, the term is used in the general characterization of predictive performance, whereas *constant bias* is used for the corresponding additive effect in method-comparison analyses.

Bias, proportional Proportional bias is a systematic error that changes with the magnitude of the measured quantity. It occurs when estimates are increasingly overestimated or underestimated as the reference value increases. In the linear regression model $x = my + b$, proportional bias is represented by deviation of the slope m from the identity slope $m = 1$.

Bonferroni Correction When conducting m simultaneous hypothesis tests at a family-wise significance level α, the Bonferroni correction sets the per-test threshold to α/m. This bounds the probability of at least one false positive, or Type I error, across the family of tests [7]. The correction is conservative when tests are positively correlated. Alternatives such as Holm's step-down procedure [24] offer greater power while maintaining family-wise error control.

Bootstrapping Bootstrapping is a resampling procedure that draws B samples of size n with replacement from the observed data to empirically approximate the sampling distribution of a statistic [8, 41]. It enables estimation of standard errors, bias, and confidence intervals without parametric distributional assumptions. It is particularly useful when the analytical distribution of a complex estimator, such as a deep learning model, is intractable.

Churn Churn, proposed by Milani Fard et al. [40], quantifies paired prediction disagreements between two models. It measures how much improvement arises from correcting previously wrong predictions rather than from net accuracy gains alone. A model may improve overall performance while still regressing on a substantial fraction of samples. Churn isolates this instability.

Clinical Effect A clinical effect is the magnitude of a difference or association deemed meaningful in a clinical or applied context, distinct from statistical significance. It is defined by domain-specific thresholds rather than p-values. A result may be statistically significant yet clinically irrelevant, or clinically meaningful yet statistically underpowered.

Coefficient of Determination (R^2) The coefficient of determination is the proportion of variance in the dependent variable explained by the model. It is defined as:

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} \in (-\infty, 1],$$

where $SS_{\text{res}} = \sum(y_i - \hat{y}i)^2$ and $SS\text{tot} = \sum(y_i - \bar{y})^2$. Values close to 1 indicate high explanatory power, whereas negative values arise when the model performs worse than a horizontal mean line. For simple linear regression through the origin, $R^2 = r^2$, where r is Pearson's correlation coefficient.

Coefficient of Variation (CV) The coefficient of variation is a dimensionless measure of relative dispersion. It expresses the standard deviation as a fraction of the mean:

$$\text{CV} = \frac{\sigma}{\mu}.$$

The CV facilitates comparison of variability across metabolites with different concentration scales. It is undefined or unstable for distributions with means close to zero.

Concise Score (r^3) The concise score is a composite scalar proposed by Rizzo et al. [46]. It is defined as the product of the ordinary least-squares regression slope b and the coefficient of determination R^2:

$$r^3 = b \cdot R^2.$$

When Pearson's r is used as the slope on standardized data, this reduces to $r \cdot r^2 = r^3$. The concise score captures both the linearity of the mapping, represented by the slope, and the explained variance, represented by R^2. This reduces the number of reported scalar metrics. Desirable values approach ± 1, and the sign indicates the direction of the relationship.

Confidence Interval (CI) A confidence interval is an interval $[\hat{\theta}_L, \hat{\theta}_U]$ constructed such that, under repeated sampling, $(1-\alpha)\times 100\%$ of such intervals contain the true parameter θ. The confidence level is a property of the procedure, not a posterior probability for a single realized interval. The latter interpretation requires Bayesian credible intervals.

Cramér–Rao Lower Bound (CRLB) The Cramér–Rao lower bound defines the theoretical minimum variance, i.e. precision, of any unbiased estimator of a scalar parameter θ [6, 9]. It is given by:

$$\mathrm{Var}(\hat{\theta}) \geq \mathcal{I}(\theta)^{-1} = \frac{\sigma^2}{\sum_{n=1}^{N} \left(\frac{\partial s[n](\theta)}{\partial \theta}\right)^2},$$

where $\mathcal{I}(\theta)$ is the Fisher information, σ^2 is the variance of the additive white Gaussian noise, and $s[n](\theta)$ is the deterministic signal sampled over $n \in N$ total data points. In MRS, fitting algorithms such as LCModel and Osprey routinely report CRLBs as percentage standard deviations, which provide per-voxel indices of quantification precision. CRLBs are lower bounds on achievable uncertainty, not estimates of actual fitting error.

Critical Region The critical region is the set of values of the test statistic for which the null hypothesis H_0 is rejected. It is constructed such that, under H_0, the probability of the test statistic falling within this region equals the significance level α.

Critical Value The critical value is the boundary of the critical region. It is the value of the test statistic beyond which H_0 is rejected at the pre-specified significance level α. The critical value is determined by the null distribution of the test statistic and the chosen α.

Cross-validation A model evaluation procedure in which the dataset is partitioned into k folds; the model is trained on $k-1$ folds and evaluated on the remaining fold, cycling through all k folds [2]. The k evaluation scores are averaged to produce a more reliable performance estimate than a single train-test split, particularly for small datasets.

Distribution A mathematical function that describes the probability of all possible outcomes of a random variable, providing a complete characterization of the process or sample.

Distributional Concordance Coefficient (C_b) The distributional concordance coefficient is the bias-correction factor in Lin's concordance correlation coefficient ρ_C. It is defined as $C_b = \rho_C / r$ [33], or equivalently as:

$$C_b = \frac{2}{\frac{\sigma_x}{\sigma_y} + \frac{\sigma_y}{\sigma_x} + \frac{(\mu_x - \mu_y)^2}{\sigma_x \sigma_y}},$$

where μ and σ are the means and standard deviations of the two samples. $C_b \in (0, 1]$, with $C_b = 1$ when the two distributions have identical means and variances. It scales down Pearson's r to penalize location and scale discordance.

Equivalence Testing Equivalence testing is a framework that inverts the conventional null and alternative hypotheses. Here, H_0 asserts that the absolute difference exceeds a prespecified equivalence margin δ, and rejection of H_0 constitutes statistical evidence of practical equivalence [48]. The standard implementation is the two one-sided tests procedure from Schuirmann [48], which requires both one-sided confidence bounds to fall within $[-\delta, +\delta]$. Equivalence testing is relevant when the goal is to demonstrate non-inferiority or interchangeability rather than a difference.

Fisher Information Fisher information quantifies how much information an observable random variable, X, carries about an unknown parameter θ. For a likelihood $p(x;\theta)$, it is defined as:

$$\mathcal{I}(\theta) = \mathbb{E}\left[\left(\frac{\partial \log p(X;\theta)}{\partial \theta}\right)^2\right]$$

Under standard smoothness and regularity conditions, meaning, where each $x \in \mathcal{X}$, the likelihood $p(x;\theta)$ is differentiable with respect to θ, this can also be written as:

$$\mathcal{I}(\theta) = -\mathbb{E}\left[\frac{\partial^2 \log p(X;\theta)}{\partial \theta^2}\right].$$

It quantifies the expected curvature of the log-likelihood with respect to θ [9], and its reciprocal is the Cramér–Rao lower bound on the variance of any unbiased estimator of θ. Higher Fisher information corresponds to greater sensitivity of the data to the parameter and tighter achievable precision.

Fit Quality Number (FQN) The fit quality number is a sample-wise scalar summary of spectral fit quality recommended by Near et al. [43]. It is derived from the ratio of the variance of fit residuals to the noise level. Descriptive statistics of the FQN over a cohort, such as the median and IQR, provide population-level characterization of fitting performance and can flag systematic model misspecification or outliers.

Heteroscedasticity Heteroscedasticity means that the residuals, or error terms, of a regression model do not have constant variance across the parameter space. In this context, heteroscedasticity permits predictive variance to change with metabolite concentration, linewidth, and SNR.

Homoscedasticity Homoscedasticity means that the residuals, or error terms, of a regression model have constant variance across the parameter space. In this context, homoscedasticity assumes constant predictive variance regardless of metabolite concentration, linewidth, and SNR.

Hypothesis Testing Hypothesis testing is a formal decision procedure for choosing between a null hypothesis H_0 and an alternative hypothesis H_1 based on observed data [44]. A test statistic is computed from the data. Its distribution under H_0 yields the p-value, which is compared to the prespecified significance level α. Hypothesis tests may target distributional location, scale, or shape, depending on the scientific question.

Hypothesis Testing, Location Location-based hypothesis tests assess differences in central tendency, such as the mean or median, between groups or against a reference value. Parametric tests include the one-sample and two-sample t-tests [18]. Non-parametric analogs include the Wilcoxon signed-rank test [61] and the Mann–Whitney U test [37] when normality cannot be assumed.

Hypothesis Testing, Scale Scale-based hypothesis tests assess differences in dispersion, such as variance or standard deviation, between groups. Common procedures include

Levene's test [32], which is robust to non-normality, Bartlett's test [3], which is sensitive to normality violations, and the F-test for equality of variances in the Gaussian setting [10].

Hypothesis Testing, Shape Shape-based hypothesis tests assess departure from a specified distributional form. Goodness-of-fit tests include the Kolmogorov–Smirnov test [28, 54], the Shapiro–Wilk test [51], which is often used for normality testing in small samples, and the Anderson–Darling test [1], which emphasizes distributional tails.

Independent and Identically Distributed (i.i.d) A fundamental concept in statistics and probability theory that describes a set of random variables or data points that share the same probability distribution, identically distributed, and do not influence each other, independent.

Intraclass Correlation Coefficient (ICC) The intraclass correlation coefficient is a reliability index derived from a variance components model, such as one-way or two-way ANOVA. It is defined as the ratio of inter-subject variance to total variance [53]:

$$\mathrm{ICC} = \frac{\sigma^2_{\text{between}}}{\sigma^2_{\text{between}} + \sigma^2_{\text{error}}}.$$

Multiple formulations exist, including ICC(1,1), ICC(2,1), and ICC(3,1), depending on whether rater effects are fixed or random and whether absolute agreement or consistency is the target. The appropriate form must be selected a priori [29]. Values ≥ 0.75 are conventionally considered good reliability [29].

Intercept of the Regression Line The intercept of the regression line is the predicted value of the dependent variable when the independent variable is zero. In an MRS quantification validation context, values close to zero indicate minimal systematic prediction offset independent of concentration [46].

Interquartile Range (IQR) The interquartile range is a robust measure of spread that describes the width of the central 50% of a distribution. It is defined as:

$$\mathrm{IQR} = Q_3 - Q_1.$$

The IQR is unaffected by extreme values. It is appropriate for non-normal error distributions and is used as the basis for the meCV and MeAPE.

Limits of Agreement (LoA) The limits of agreement were introduced by Bland and Altman [5] for method comparison studies. The 95% limits of agreement are defined as:

$$\bar{d} \pm 1.96 s_d,$$

where $\bar{d}$ and s_d are the mean and standard deviation of pairwise differences, respectively. They define the interval within which 95% of individual differences are expected to lie, given that the differences are approximately normally distributed. A Bland–Altman plot of d_i against the mean of the paired measurements, $(\hat{y}_i + y_i)/2$, should be inspected for proportional bias.

Lin's Concordance Correlation Coefficient (ρ_C) Lin's concordance correlation coefficient measures agreement between two continuous variables by simultaneously penalizing deviations from perfect correlation and from the line of identity $y = x$ [33]. It is defined as:

$$\rho_C = \frac{2\sigma_{xy}}{\sigma_x^2 + \sigma_y^2 + (\mu_x - \mu_y)^2} = C_b \cdot r \in [-1, 1].$$

It decomposes into Pearson's r, which measures precision, scaled by C_b, which measures accuracy. $\rho_C = 1$ requires both $r = 1$ and $C_b = 1$.

Marginal Distribution The marginal distribution is the distribution of one or more variables obtained by integrating out all remaining variables from a joint distribution. For example, for a single variable the marginal distribution is just $p(x)$, the probability of x. However for two variables x and y, it is given by

$$p(x) = \int p(x, y), \mathrm{d}y, \tag{4}$$

where $p(x)$ is the marginal probability of x given the joint probability of x and y, and for three variables, x, y, and z, where the marginal distribution of x becomes

$$p(x) = \int \int p(x, y, z), \mathrm{d}y\mathrm{d}z, \tag{5}$$

where only the variable of interest is preserved. In Bayesian predictive modeling, the marginal predictive distribution is obtained by integrating over the posterior distribution of model parameters:

$$p(y^*|\mathbf{x}^*, \mathcal{D}) = \int p(y|\mathbf{x}^*, \boldsymbol{\theta}), p(\boldsymbol{\theta}|\mathcal{D}), \mathrm{d}\boldsymbol{\theta}. \tag{6}$$

This integrates predictions across the full posterior distribution, rather than relying on a single point estimate.

Mean Absolute Percent Error (MAPE) The mean absolute percent error measures the average magnitude of the relative prediction error. It is defined as:

$$\mathrm{MAPE} = \frac{1}{n} \sum_{i=1}^{n} \left| \frac{\hat{y}_i - y_i}{y_i} \right| \times 100\%.$$

MAPE is symmetric about zero error when residuals are unbiased [36]. It is useful in MRS quantification because the 15% clinical threshold is expressed as a relative quantity, but it is undefined for zero-valued ground truth values.

Mean (Population, μ) The population mean is the first moment of a distribution. It is defined as $\mu = \mathbb{E}[X] = \int x, p(x), \mathrm{d}x$. It characterizes the location of the distribution and is equal to the arithmetic average in the discrete, uniform case.

Mean (Sample, $\bar{x}$) The sample mean is the arithmetic average of the observed values. It is defined as $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$. Under random sampling, it is an unbiased and consistent estimator of the population mean μ.

Mean Percent Error (MPE) The mean percent error is a signed relative error metric that quantifies systematic bias. It is defined as:

$$\mathrm{MPE} = \frac{1}{n} \sum_{i=1}^{n} \frac{\hat{y}_i - y_i}{y_i} \times 100\%.$$

MPE distinguishes systematic overestimation, where MPE > 0, from underestimation, where MPE < 0. It is appropriate for deriving mean correction factors when directional bias is consistent across the concentration range.

Median Absolute Percent Error (MeAPE) The median absolute percent error is the median of $\left|\frac{\hat{y}_i - y_i}{y_i}\right| \times 100\%$. It is a robust, non-parametric counterpart to MAPE, insensitive to outliers and suitable for skewed error distributions. MeAPE is preferable to MAPE when even a small number of extreme errors would otherwise dominate the mean.

Median Coefficient of Variation (meCV) The median coefficient of variation is a robust analog of CV for non-normal distributions. It expresses relative spread without sensitivity to outliers or distributional skew and is defined as:

$$\text{meCV} = \frac{Q_3 - Q_1}{\tilde{x}} = \frac{\text{IQR}}{\text{median}}.$$

Median Percent Error The median percent error is the median of $\frac{\hat{y}_i - y_i}{y_i} \times 100\%$. It is a robust measure of signed systematic bias appropriate when the error distribution is non-normal or contains outliers. It is interpreted analogously to MPE but is minimized by the conditional median rather than the conditional mean.

Median Root Squared Error (MeRSE) The median root squared error is the median of $|\hat{y}_i - y_i|$, that is, the median absolute error. It is a robust alternative to RMSE that down-weights large individual errors. MeRSE is minimized by the conditional median, unlike RMSE, which is minimized by the conditional mean. This makes MeRSE more informative when the residual distribution is heavy-tailed.

Metabolite SNR (mSNR) Metabolite-level signal-to-noise ratio is calculated either using a spectral or power formulation. The spectral SNR is defined as the maximal peak height of the real spectral component divided by the standard deviation of a noise region. The power-based SNR is defined as the square root of the power of the metabolite signal divided by one standard deviation of the noise floor. The power can be calculated by integrating the spectrum (FD) or by taking the first FID point (TD). mSNR is a per-metabolite quantity, in contrast to sample-level SNR, which is typically referenced to a single resonance, such as water or total creatine, or total power.

Mixed Effects Model A mixed effects model is a regression framework that incorporates fixed effects, which are population-level parameters shared across all subjects, and random effects, which are subject- or group-specific deviations drawn from a distribution [45]. It enables valid inference on repeated measures and hierarchically structured data by explicitly modeling within-subject correlation. Random effects are typically assumed to be Gaussian with zero mean.

Negative Log Predictive Density (NLPD) The negative log predictive density is a strictly proper scoring rule that evaluates the full predictive distribution [14]. For observations y_i and predictive densities $p(y_i \mid \mathbf{x}i)$, it is defined as:

$$\text{NLPD} = -\frac{1}{n}\sum i = 1^n \log p(y_i \mid \mathbf{x}_i).$$

NLPD simultaneously penalizes inaccurate mean predictions and poorly calibrated uncertainty. Lower values indicate better-calibrated probabilistic models. Unlike point-error metrics, NLPD rewards well-characterized predictive variance.

Normalized Root Mean Squared Error (NRMSE) The normalized root mean squared error rescales RMSE to enable comparison across measurement scales. It is commonly defined as:

$$\text{NRMSE} = \frac{\text{RMSE}}{y_{\max} - y_{\min}},$$

although normalization by the mean or standard deviation of y is also used. The normalization convention must be stated explicitly when comparing across studies.

Pearson's Correlation Coefficient (r) Pearson's correlation coefficient measures the strength and direction of the linear relationship between two variables. It is defined as:

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}} \in [-1, 1].$$

It is equivalent to the OLS regression slope when both variables are standardized to zero mean and unit variance. Pearson's r is sensitive to outliers and does not capture agreement, only association.

Permutation Testing Permutation testing is a non-parametric procedure that generates the null distribution of a test statistic by recomputing it under P random permutations of the observed labels [15]. The p-value is the fraction of permuted statistics at least as extreme as the observed value. The procedure is valid under exchangeability of observations under H_0 and does not require distributional assumptions.

Prediction Interval A prediction interval is an interval $[\hat{y}_L, \hat{y}_U]$ designed to contain a single future observation with probability $(1 - \alpha) \times 100\%$. Prediction intervals are wider than confidence intervals because they account for both parameter uncertainty and irreducible observation noise. They are appropriate when the goal is to bound individual predictions rather than population parameters.

Predictive Entropy Predictive entropy quantifies total uncertainty in a probabilistic model's prediction. For a predictive distribution $p(y \mid \mathbf{x})$, it is defined as:

$$H[y \mid \mathbf{x}] = -\int p(y \mid \mathbf{x}) \log p(y \mid \mathbf{x}) \, \mathrm{d}y.$$

Predictive entropy encompasses both aleatoric uncertainty and epistemic uncertainty, which reflects model predictive uncertainty [11]. High predictive entropy identifies inputs for which the model is uncertain and can be useful as one criterion for detecting unreliable or out-of-distribution predictions.

Predictive Marginal Variance The predictive marginal variance is the variance of the predictive distribution $p(y \mid \mathbf{x}, \mathcal{D})$ after marginalizing over model parameters. Using the law of total variance, it can be decomposed as:

$$\mathbb{V}[y \mid \mathbf{x}, \mathcal{D}] = \mathbb{E}_{\boldsymbol{\theta} \sim p(\boldsymbol{\theta} \mid \mathcal{D})} \left[\mathbb{V}[y \mid \mathbf{x}, \boldsymbol{\theta}]\right] + \mathbb{V}_{\boldsymbol{\theta} \sim p(\boldsymbol{\theta} \mid \mathcal{D})} \left[\mathbb{E}[y \mid \mathbf{x}, \boldsymbol{\theta}]\right].$$

The first term reflects aleatoric uncertainty, while the second reflects epistemic uncertainty. In MC-dropout implementations, Gal estimates predictive variance empirically from stochastic forward passes, with an additional observation-noise term [11].

Residuals, Orthogonal Orthogonal residuals are perpendicular distances from observed data points to the fitted line or surface, as used in orthogonal, or total, least-squares regression [38]. Unlike the vertical residuals of OLS, orthogonal residuals account for measurement error in both

variables and minimize the sum of squared perpendicular distances. They are appropriate when neither variable can be treated as error-free, as is common in method comparison studies.

Root Mean Squared Error (RMSE) The root mean squared error measures the typical magnitude of prediction errors in the same units as the target variable. It is defined as:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2}.$$

RMSE penalizes large errors quadratically. In MRS quantification, RMSE is expected to be on the order of the CRLBs for a well-fitted model [46].

Sample Size (n) Sample size is the number of independent observations in a study. It governs statistical power and the precision of parameter estimates. Formal *a priori* power analyses specify the sample size n required to detect a given effect size δ at chosen Type I error rate α and Type II error rate β. Post hoc power analyses are generally discouraged because they are redundant with the obtained p-value.

Significance Level (α) The significance level is the pre-specified probability of committing a Type I error. It defines the threshold below which p-values lead to rejection of H_0. The significance level must be chosen before data collection. Conventional values of 0.05 and 0.01 are starting points, and domain-appropriate justification is preferable.

Slope of the Regression Line The slope of the regression line is the rate of change of the predicted variable per unit change in the ground truth. In an MRS quantification validation study, slopes close to 1.0 indicate proportionally correct mapping across the full concentration range, with no differential underestimation or overestimation at high versus low concentrations [46].

Signal-to-Noise Ratio (SNR) The signal-to-noise ratio is the ratio of signal amplitude or power to noise amplitude or power. In MRS, it is commonly defined as the peak height of a reference resonance divided by twice the standard deviation of a signal-free spectral region. SNR is the primary determinant of quantification precision and directly governs the achievable CRLBs for each metabolite.

Statistical Significance Statistical significance is determined by the probability of observing a test statistic at least as extreme as the one obtained, assuming H_0 is true. This probability is the p-value. Statistical significance does not quantify effect magnitude, clinical relevance, or reproducibility, and it should be interpreted alongside effect size estimates and confidence intervals.

Symmetric Mean Absolute Percent Error (sMAPE) The symmetric mean absolute percent error was proposed to address a perceived asymmetry in MAPE [36]. It is defined as:

$$\mathrm{sMAPE} = \frac{1}{n}\sum_{i=1}^{n}\frac{|\hat{y}_i - y_i|}{(|y_i| + |\hat{y}_i|)/2} \times 100\%.$$

However, sMAPE itself introduces asymmetry between overestimation and underestimation and is undefined when both y_i and $\hat{y}_i$ are zero [17]. Its use is generally not recommended. The perceived asymmetry in MAPE does not occur when errors are centered at zero.

Test Statistic A test statistic is a scalar function of the observed data whose distribution under H_0 is known analytically or by simulation. It is used to compute the p-value or to compare against a critical value. Examples include the t-statistic, F-statistic, and χ^2-statistic.

Theil–Sen Estimator The Theil–Sen estimator is a robust non-parametric slope estimator computed as the median of all $\binom{n}{2}$ pairwise slopes [57, 49]. It is defined as:

$$\hat{b}\mathrm{TS} = \mathrm{med}(i<j)\left(\frac{y_j - y_i}{x_j - x_i}\right).$$

It achieves a breakdown point of $\approx 29.3\%$ and is appropriate when residuals are non-normal or heteroscedastic. It is more robust than OLS in the presence of outlying concentration estimates.

Tolerance Interval A tolerance interval is an interval $[L, U]$ designed to contain at least a specified proportion p of the population with confidence $1-\alpha$ [21]. It is distinct from a prediction interval, which targets a single future observation, and from a confidence interval, which targets a parameter. Tolerance intervals make a joint probability statement about a fraction of the distribution, making them suitable for defining acceptable ranges of quantification error across a clinical population.

Type I Error (α) A Type I error occurs when H_0 is rejected although it is true. It is a false positive, and its probability is controlled by the significance level α. In the context of multiple comparisons, the family-wise Type I error rate increases with the number of tests unless corrected by procedures such as the Bonferroni or Holm correction.

Type II Error (β) A Type II error occurs when H_0 is not rejected although it is false. It is a false negative, and its probability is β. Statistical power is defined as $1-\beta$ and is controlled by study design through sample size, effect size, measurement precision, and significance level.

Variance Characterizes the spread of a distribution. In statistics, it is the expected squared deviation of a random variable from its mean: $\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2]$. Refer to Appendix A for the deep learning definition.